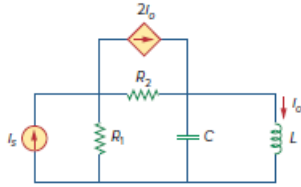


Problem

Using Fig. 10.61, design a problem to help other students better understand nodal analysis.

Figure 10.61:



Step-by-step solution

Step 1 of 5

Refer to Figure 10.61 in the textbook.

Draw the frequency domain circuit as shown in Figure 1.

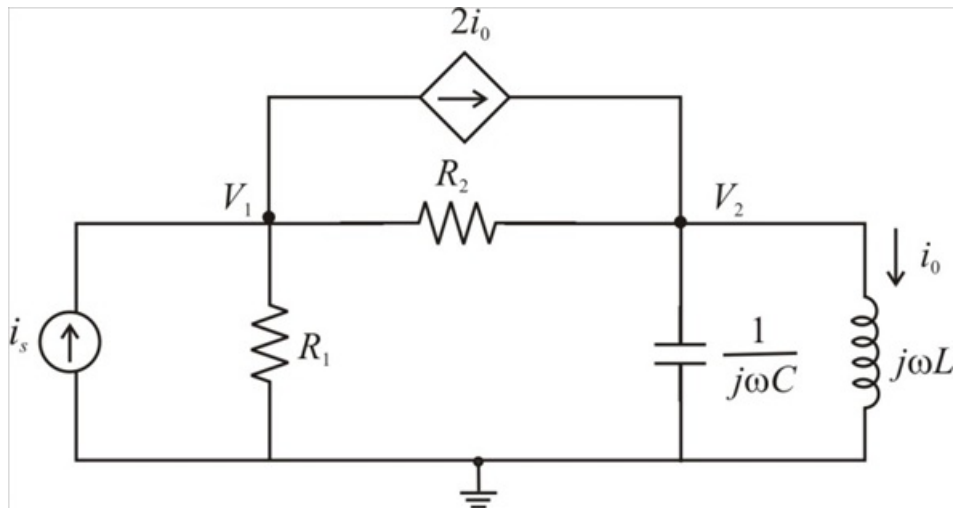


Figure 1

Step 2 of 5

From Figure 1, write the equation for i_o .

$$i_o = \frac{V_2}{j\omega L} \quad \dots\dots (1)$$

Apply Kirchhoff's current law at node V_1 .

$$i_s = \frac{V_1}{R_1} + 2i_o + \frac{V_1 - V_2}{R_2}$$

$$i_s = \frac{V_1 R_2 + 2i_o R_1 R_2 + V_1 R_1 - V_2 R_1}{R_1 R_2}$$

$$i_s R_1 R_2 = V_1 (R_1 + R_2) - V_2 R_1 + 2i_o R_1 R_2$$

$$V_1 (R_1 + R_2) - V_2 R_1 + 2i_o R_1 R_2 - i_s R_1 R_2 = 0$$

$$V_1 (R_1 + R_2) - V_2 R_1 + R_1 R_2 (2i_o - i_s) = 0$$

$$V_1 = \frac{V_2 R_1 - R_1 R_2 (2i_o - i_s)}{R_1 + R_2} \quad \dots\dots (2)$$

Apply Kirchhoff's current law at node V_2 .

$$2i_0 = i_0 + \frac{V_2 - V_1}{R_2} + \frac{V_2}{\frac{1}{j\omega C}}$$

$$i_0 - 2i_0 + \frac{V_2 - V_1}{R_2} + j\omega CV_2 = 0$$

$$-i_0 + \frac{V_2 - V_1}{R_2} + j\omega CV_2 = 0$$

$$\frac{-i_0 R_2 + V_2 - V_1 + j\omega CR_2 V_2}{R_2} = 0$$

$$-V_1 + (1 + j\omega CR_2)V_2 - i_0 R_2 = 0 \dots\dots (3)$$

Step 4 of 5

Replace equation 2 in equation 3.

$$-\left[\frac{V_2 R_1 - R_1 R_2 (2i_0 - i_s)}{R_1 + R_2} \right] + (1 + j\omega CR_2)V_2 - i_0 R_2 = 0$$

$$\frac{-V_2 R_1 + R_1 R_2 (2i_0 - i_s) + (1 + j\omega CR_2)(R_1 + R_2)V_2 - i_0 R_2 (R_1 + R_2)}{R_1 + R_2} = 0$$

$$(-R_1 + (1 + j\omega CR_2)(R_1 + R_2))V_2 + 2R_1 R_2 i_0 - R_1 R_2 i_s - R_1 R_2 i_0 - R_2^2 i_0 = 0$$

$$(-R_1 + (1 + j\omega CR_2)(R_1 + R_2))V_2 + i_0 R_1 R_2 - i_0 R_2^2 - i_s R_1 R_2 = 0$$

$$(-R_1 + (1 + j\omega CR_2)(R_1 + R_2))V_2 = -i_0 R_1 R_2 + i_0 R_2^2 + i_s R_1 R_2$$

$$V_2 = \frac{i_0 R_2 (R_2 - R_1) + i_s R_1 R_2}{[-R_1 + (1 + j\omega CR_2)(R_1 + R_2)]} \dots\dots (4)$$

Step 5 of 5

Determine the current i_o .

Replace equation 4 in equation 1.

$$i_0 = \frac{\left[\frac{i_0 R_2 (R_2 - R_1) + i_s R_1 R_2}{[-R_1 + (1 + j\omega CR_2)(R_1 + R_2)]} \right]}{j\omega L}$$

$$i_0 = \frac{i_0 R_2 (R_2 - R_1) + i_s R_1 R_2}{-j\omega LR_1 + (1 + j\omega CR_2)j\omega L(R_1 + R_2)}$$

$$-i_0 j\omega LR_1 + i_0 (1 + j\omega CR_2)(j\omega LR_1 + j\omega LR_2) = i_0 R_2 (R_2 - R_1) + i_s R_1 R_2$$

$$-i_0 j\omega LR_1 + i_0 (j\omega LR_1 + j\omega LR_2 + j^2 \omega^2 LCR_1 R_2 + j^2 \omega^2 LCR_2^2) - i_0 R_2^2 + i_0 R_1 R_2 = i_s R_1 R_2$$

$$-i_0 j\omega LR_1 + i_0 (j\omega LR_1 + j\omega LR_2 - \omega^2 LCR_1 R_2 - \omega^2 LCR_2^2) - i_0 R_2^2 + i_0 R_1 R_2 = i_s R_1 R_2$$

$$-i_0 (j\omega LR_1 - j\omega LR_1 - j\omega LR_2 + \omega^2 LCR_1 R_2 + \omega^2 LCR_2^2 + R_2^2 - R_1 R_2) = i_s R_1 R_2$$

$$i_0 = -i_s \times \left[\frac{R_1 R_2}{\omega^2 LCR_2 (R_1 + R_2) - j\omega LR_2 + R_2^2 - R_1 R_2} \right]$$

Therefore, the current i_o is $\boxed{-i_s \times \left[\frac{R_1 R_2}{\omega^2 LCR_2 (R_1 + R_2) - j\omega LR_2 + R_2^2 - R_1 R_2} \right] \text{ A}}.$